

MicroSense AI-Cam Report

AI-assisted optical monitoring report for preliminary screening of microplastic-like particle candidates in water samples.

1. Sample Summary

Report Generated	08 Jun 2026, 10:17 AM
Sample ID	14
Sample Source	Tap Water
Chamber Volume	50 ml
File Type	image
Monitoring Risk Level	Low

2. Detection and Monitoring Results

Detected Candidates	5
Estimated Particles / Litre	100
MSMI Score	16
MPI Score	16
Confidence Score	75
Particle Size Category	100–500 µm estimated (prototype)
Average Particle Area	1053.60
Average Brightness	42.35

3. Hybrid Validation Results

Raw Detection Count	5
Accepted Candidate Count	5
Rejected Candidate Count	0
Hybrid Validation Score	76.05
Filter Summary	All 5 detected particle candidates passed the hybrid validation filter.

4. Image Quality Assessment

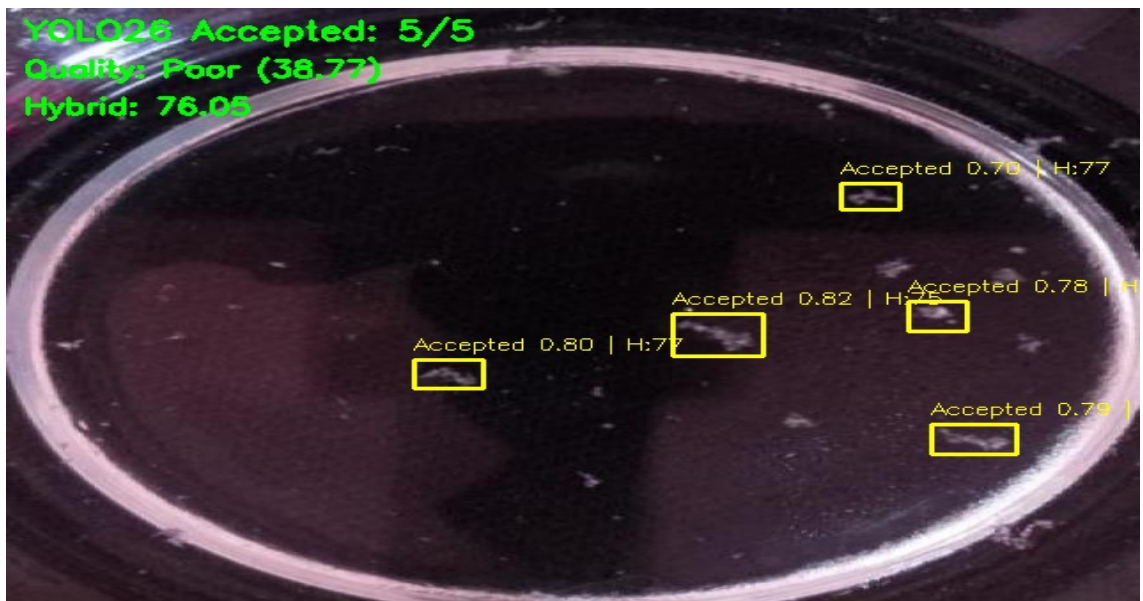
Image Quality Status	Poor
Image Quality Score	38.77
Focus Score	28.94
Brightness Score	33.17
Contrast Score	56.18
Overexposed Percent	0.43
Underexposed Percent	42.95
Quality Warning	Image may be blurry. Image is too dark. High underexposure detected.

5. Risk Interpretation

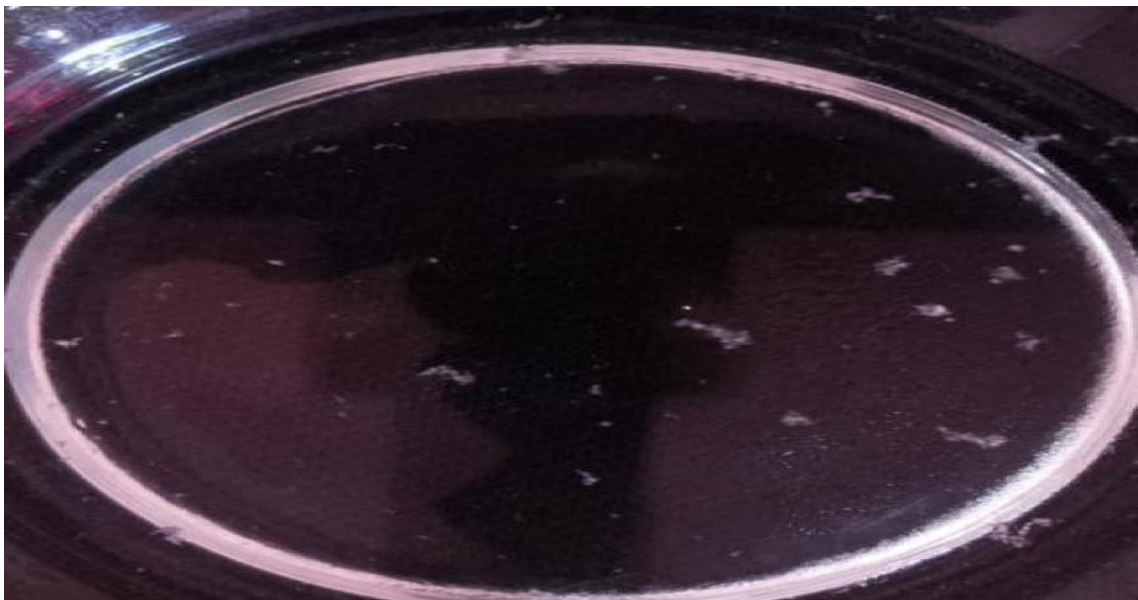
Source Risk Factor	1
Concentration Score	50
Size Score	40
Concentration-Only Risk Level	Moderate
Risk Explanation	MSMI is calculated using estimated particle concentration, detected count, average particle size, and particle density.
Recommendation	Low monitoring concern. Continue normal monitoring and retest periodically.
Notes	Analyzed using YOLO26n main detector

6. Image Evidence

Processed Output Image



Original Uploaded Image



7. Scientific Scope and Disclaimer

This report is generated by MicroSense AI-Cam, a prototype-level AI-assisted optical monitoring system. The reported objects represent microplastic-like particle candidates identified from image-based visual features. The system does not chemically confirm polymer composition and does not replace certified laboratory techniques such as FTIR spectroscopy, Raman spectroscopy, or advanced microscopic analysis. The generated values, including estimated concentration, MSMI score, confidence score, and monitoring risk level, should be interpreted only as preliminary screening indicators and not as regulatory-grade or laboratory-certified measurements.

8. System Pipeline

Water sample image acquisition → YOLO26n-based candidate detection → OpenCV-based fallback support where required → hybrid visual validation → image quality assessment → monitoring score calculation → database storage → dashboard and report generation.